Lab 6.1 – Create an Input

At the end of this lab exercise, Participants will be able to:

1. Advance through the sequence to create an input

The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

2. Easy Builder User Interface Application Step 2
    1. Inputs

Follow the steps below to complete the lab exercise:

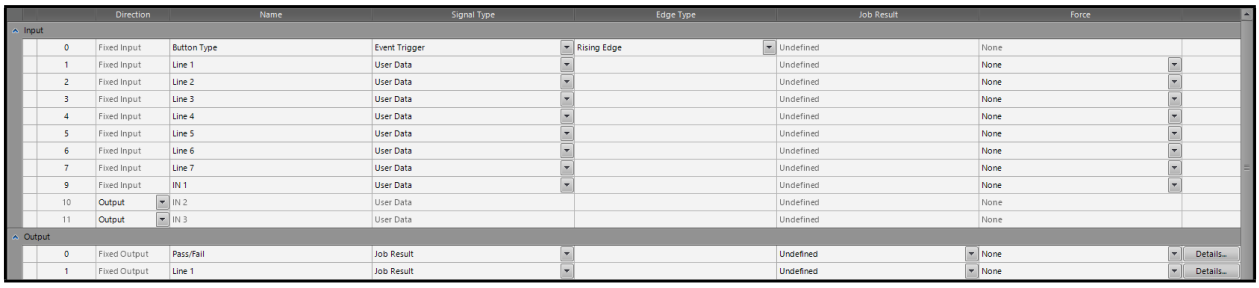
1. Continue with the **MyFilter** job from the previous lab.

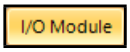
   NOTE: *Confirm the demo plate is on the PASS side in the image window*.

2. Click the **Inputs/Outputs** 

Application Step button.

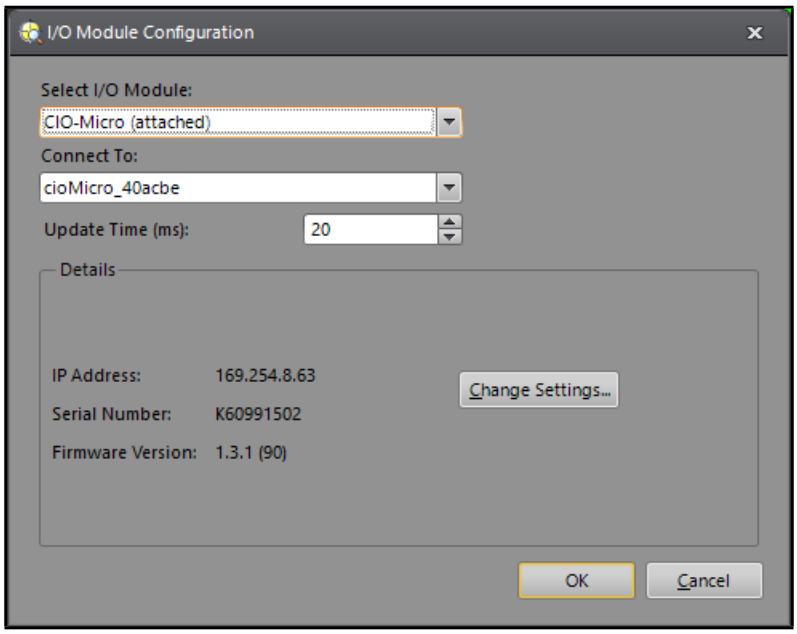
   The bottom of the UI – Settings pane will be replaced with I/O boxes.



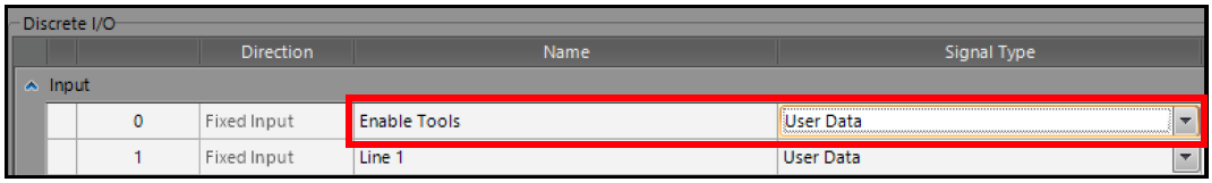
3. If the needed lines are not available (greyed out) click the **I/O Module** 

button at the bottom left.

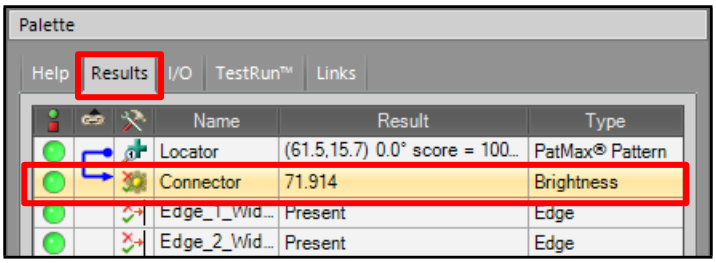
   The **I/O Module Configuration** box displays.

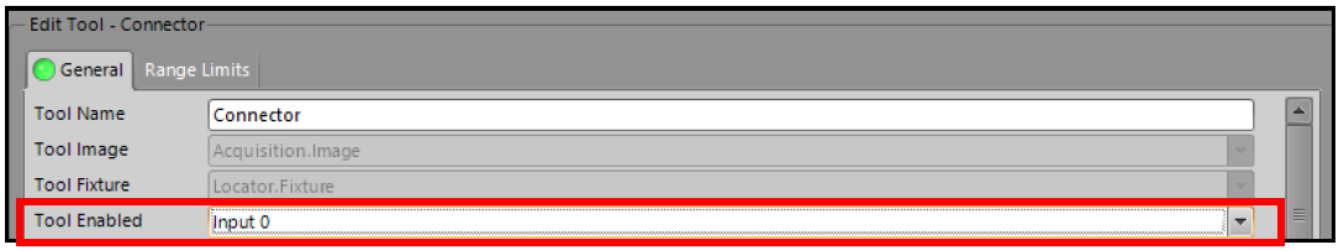


4. Select the **CIO-Micro (attached)** or the I/O module being used in lab and click the **OK** OK button.
5. Rename Line 0 **Enable Tools** and confirm the Signal Type is set to **User Data**.



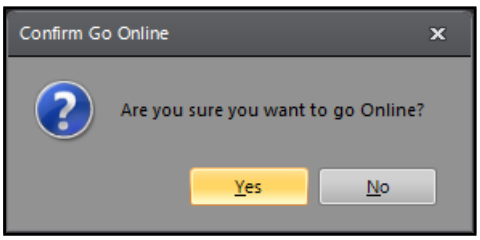
6. On the **Results** tab of the Palette double-click the Connector tool to open and change the **Tool Enabled** parameter from On to **Input 0.**





7. Click the **Online** button in the Toolbar.

   The **Confirm Go Online** dialog displays. Click the Yes Yes button.



8. Confirm the camera is **Online** Online in the lower right-hand corner below the Filmstrip.

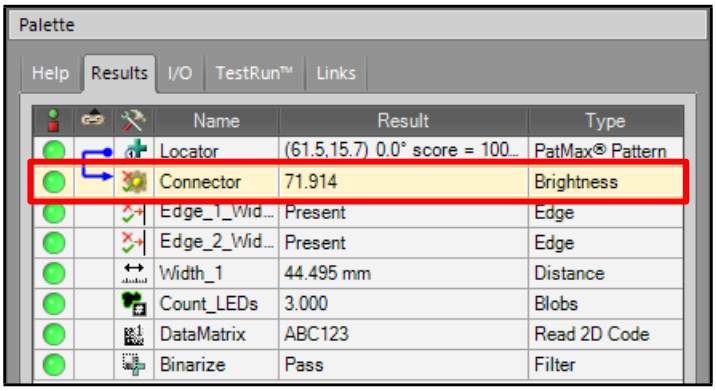


9. Click the button that is wired to Input 0 of the I/O module.

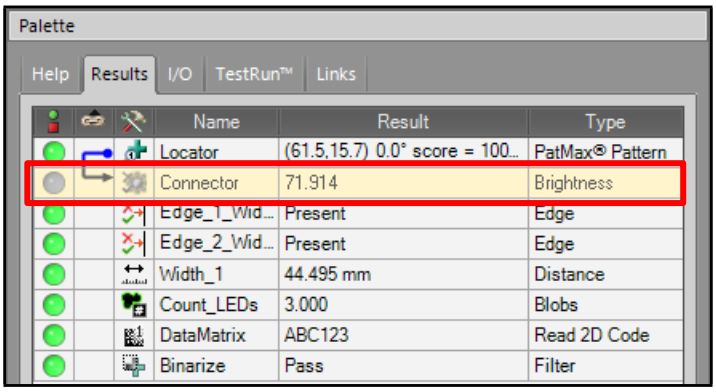


NOTE: *The Connector tool will be enabled or disabled based on Input 0 when an image is acquired. Input 0 enables the tool, but it does not execute until an image is acquired.*

Button Pressed



Button Released



10. Go offline. Save the job as **MyInput** on the In-Sight camera and in your own folder on the PC.

Lab 6.2 – Create an Output

At the end of this lab exercise, Participants will be able to:

1. Advance through the sequence of tools to final output

The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

2. Easy Builder User Interface Application Step 2
    1. Outputs

Follow the steps below to complete the lab exercise:

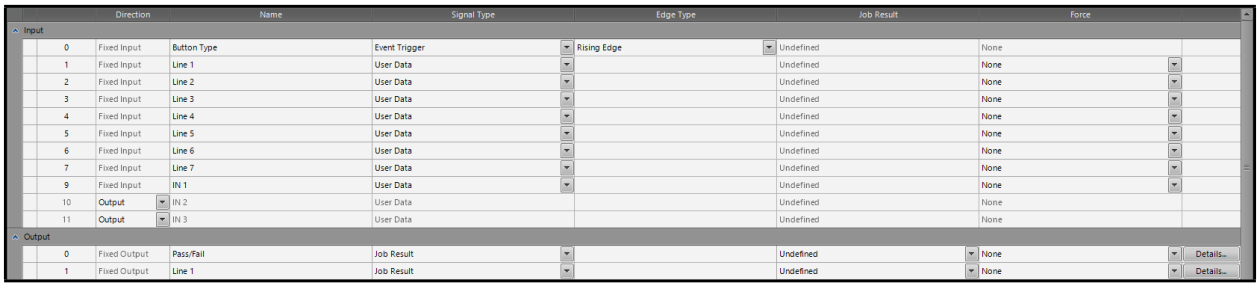
1. Continue with the **MyInput** job from the previous lab.

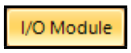
   NOTE: *Confirm the demo plate is on the PASS side in the image window.*

2. Continue the Inputs / Outputs 

Application Step.

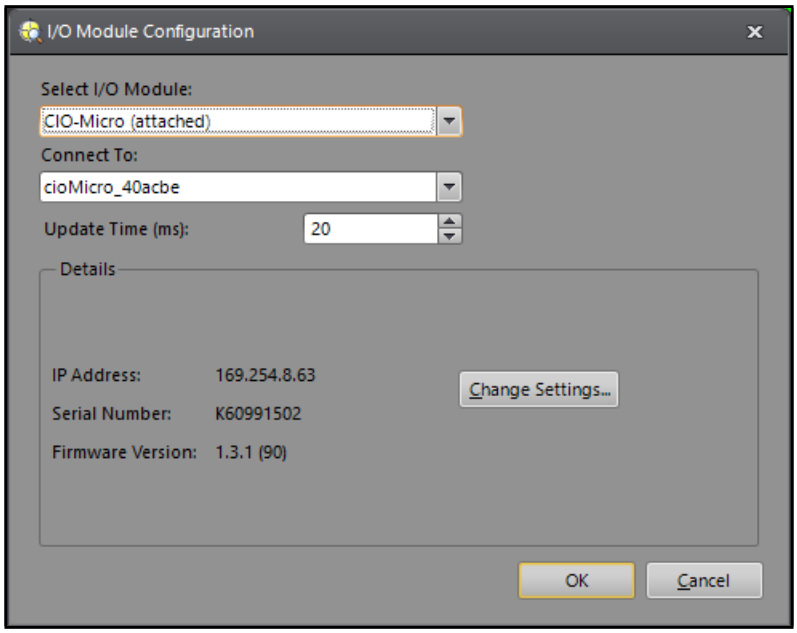
   The bottom of the UI – Settings pane will be replaced with I/O boxes.



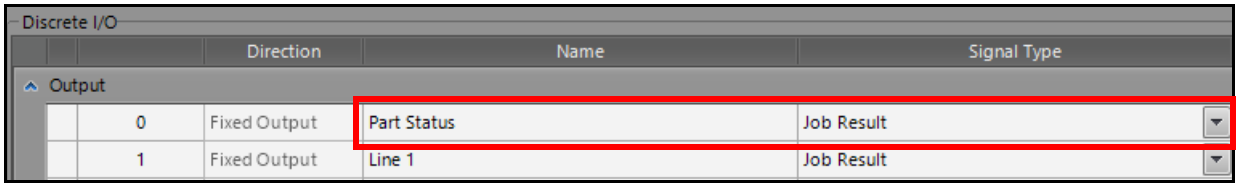
3. If the needed lines are not available (greyed out) click the **I/O Module** 

button at the bottom left

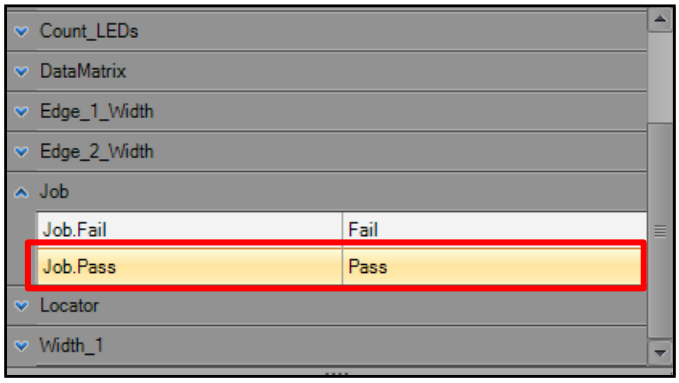
   The **I/O Module Configuration** box displays.

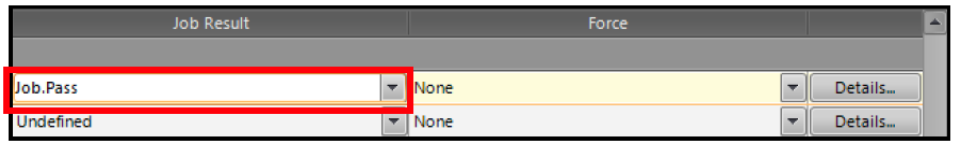


4. Select the **CIO-Micro (attached)** or the IO module being used in the lab and click the **OK** OK button.
5. Scroll to the Output section of the Discrete I/O section, rename Line 0 **Part Status** and confirm the Signal Type is set to **Job Result**.



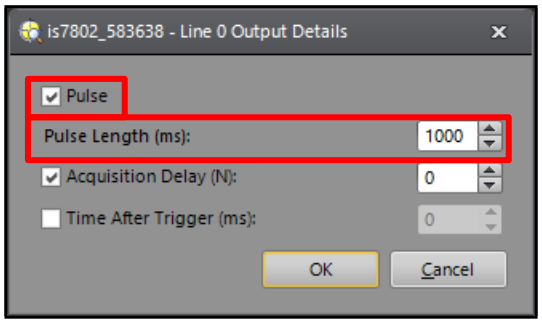
6. Change the **Job Result** parameter from Undefined to **Job.Pass**.





7. Click the **Details** Details... button, the Line 0 Output Details dialog displays.

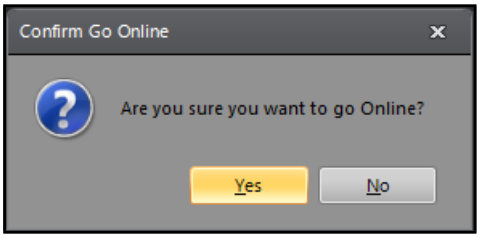
   Check the **Pulse** checkbox and enter **1000** in the Pulse Length box.



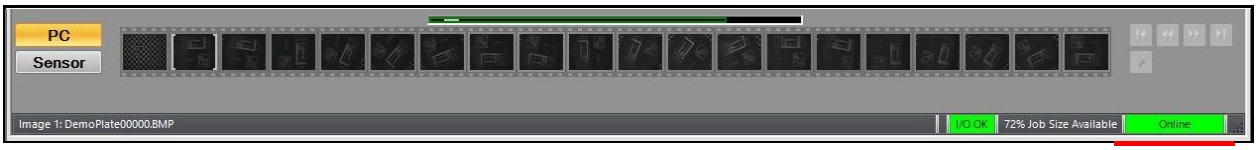
NOTE: *Enter 1000 ms for the lab – this is so that you can see the LED light on the I/O Module. A PLC could handle much less.*

8. Click the **Online** button in the Toolbar.

The **Confirm Go Online** dialog displays. Click the Yes button.



9. Confirm the camera is **Online** in the lower right-hand corner below the Filmstrip. Do a manual trigger with the F5 keyboard key, while watching the LED on the I/O Expansion Module. (Try to look directly down on the LED, because it is hard to see at an angle.)



NOTE: *The Output at Line 0 on the I/O expansion module pulses when the part is passing.*

10. Go offline. Save the job as **MyOutput** on the In-Sight camera and in your own folder on the PC.

Lab 6.3 – Communication

At the end of this lab exercise, Participants will be able to:

1. Set up an FTP and send data to the designated folder

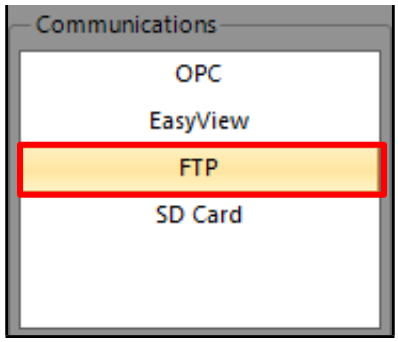
The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

2. Easy Builder User Interface Application Step 3
    1. Communication

Follow the steps below to complete the lab exercise:

1. Load the **MyFilter** job from the previous lab.
   NOTE: Confirm the demo plate is on the PASS side in the image window.
2. Save the job as **MyFTP** in the folder created in Lab #2.
3. Create a new folder named **EZB_FTP** on your PC. Copy the path of this folder.
   NOTE: This folder will be the Authorized FTP Directory where your images will be saved.
4. Go to the **Systems** menu and select **Options** → **Emulation** and paste this path into the **Authorized FTP Directory** field.
5. Click the **Communication** 

Application Step button.
6. From the Communications window select **FTP**.



The **FTP** dialog displays.

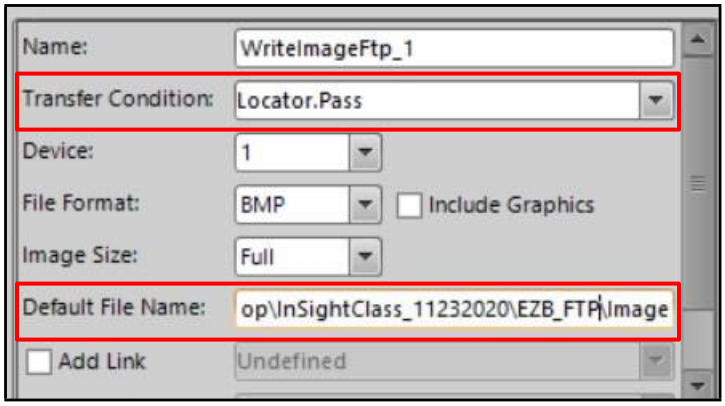


7. Click the **Add** button.

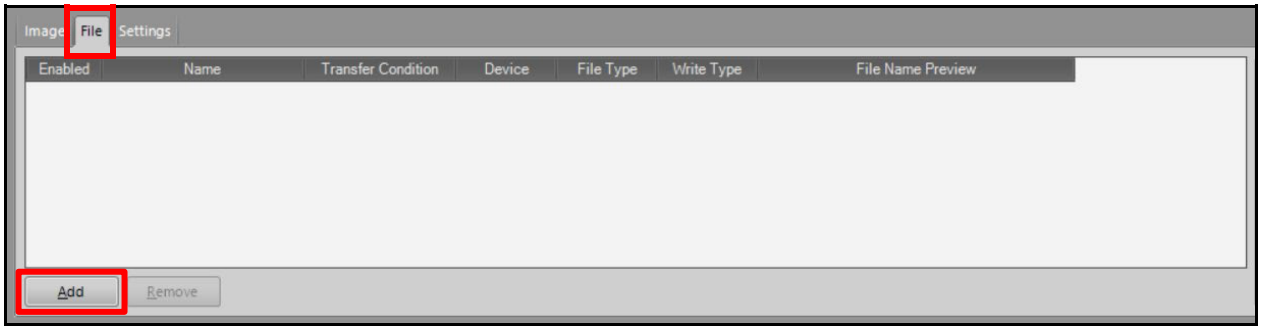
A **WriteImageFTP** tool is added.



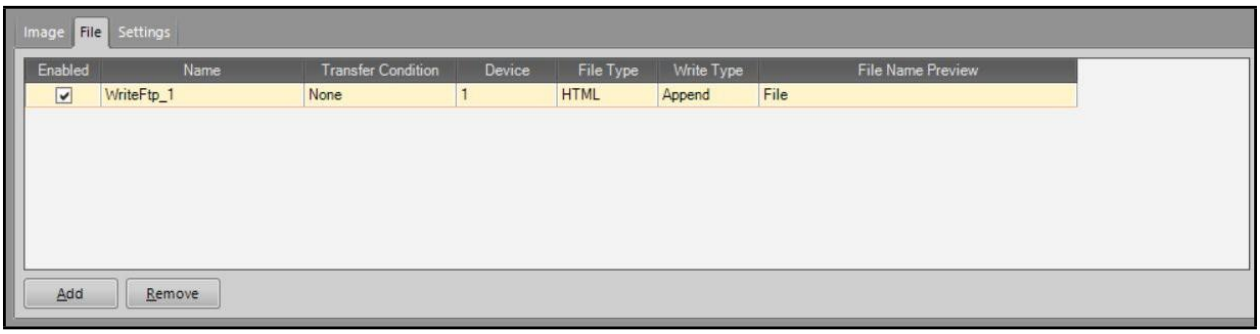
8. Set the parameters for the **WriteImageFTP** as follows:
   - **Transfer Condition**: *Locator.Pass*
   - **Default File Name**: paste the *file path* in front of the word **\Image**
   - Allow the defaults for the remaining parameters



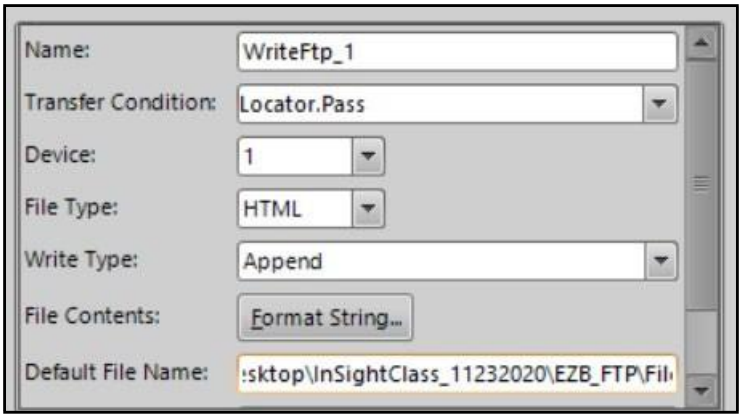
9. Click the **File** tab and click the **Add** button.



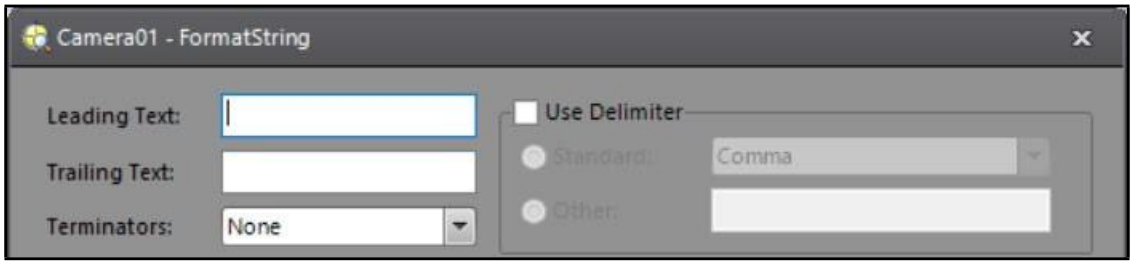
A **WriteFTP** tool is added.



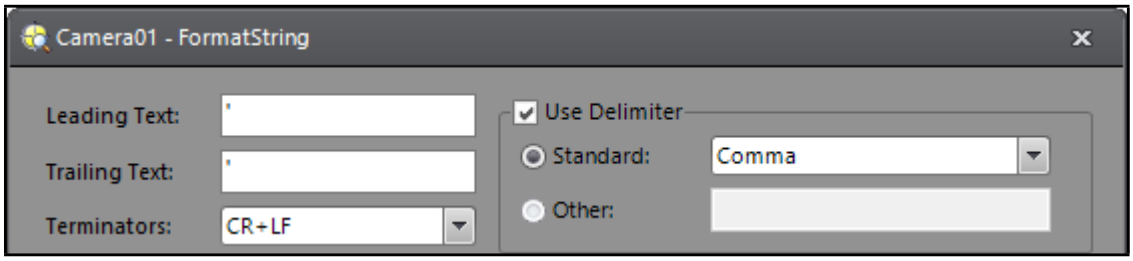
10. Set the parameters for the **WriteFTP** as follows:
    - Transfer Condition: *Locator.Pass*
    - Default File Name: paste the *file path* in front of the word **\File**
    - Allow the defaults for the remaining parameters



11. Click the **Format String** button.
    The **Format String** dialog displays.

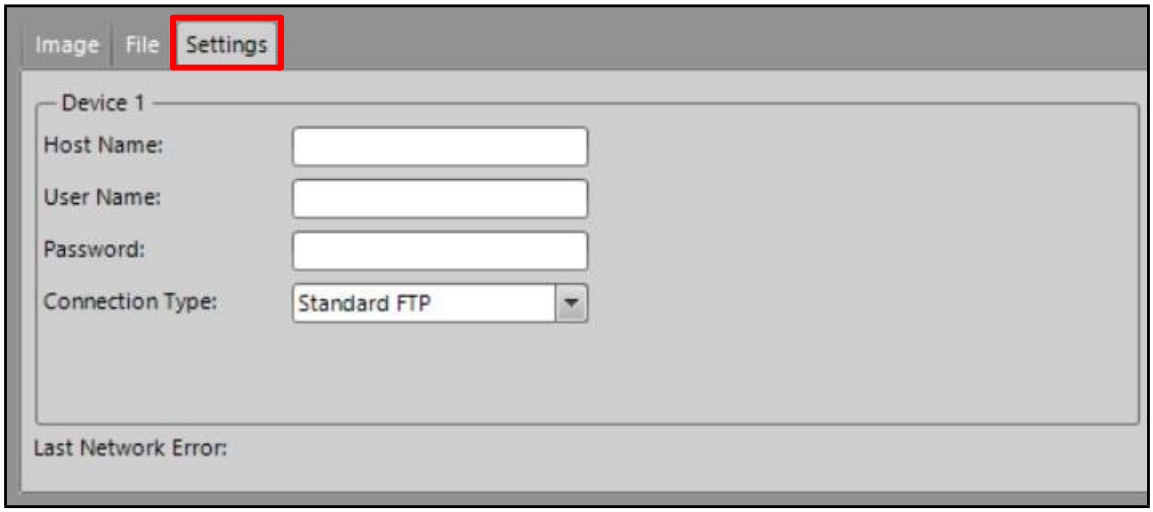


12. Format the String as follows:
    - Leading Text: ‘ (single quote)
    - Trailing Text: ‘ (single quote)
    - Terminators: CR+LF
    - Use Delimiter: check the checkbox

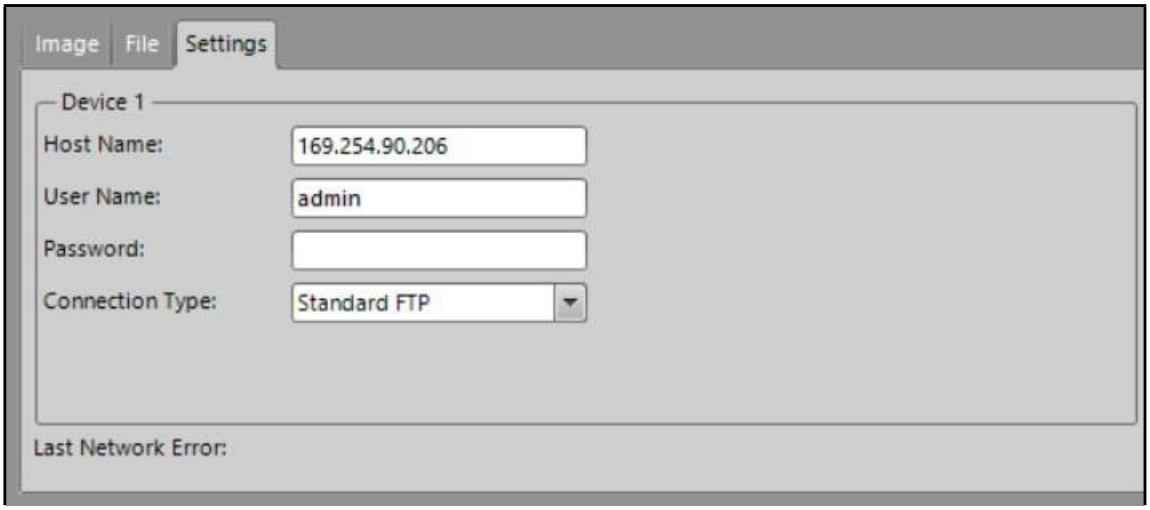


13. Click the **OK** button to close the FormatString dialog.

14. Click the **Settings** tab.
    The **Device** dialog displays.

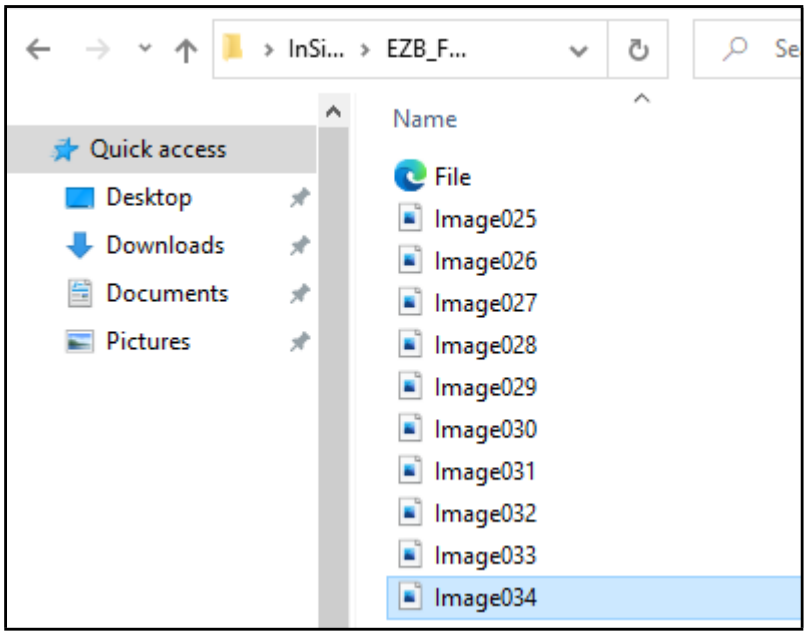


15. Enter the **IP Address** of your PC in the Host Name field.
16. Enter **admin** as the User Name: and leave the Password field blank.



17. Go **Online**.
18. Click the **Trigger** button to acquire images. For each acquired image, a bitmap file is stored in the Authorized FTP Directory.
19. Once you have acquired several images go **Offline**.

20. Open the **EZB_FTP** folder and review the results.



21. Save your job.

Lab 6.4 – Deployment

At the end of this lab exercise, Participants will be able to:

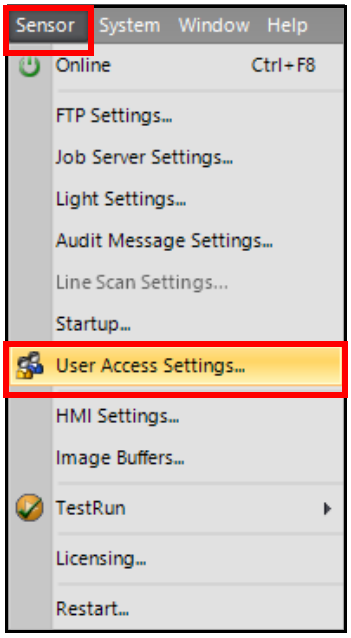
1. Utilize the utilities available in EasyBuilder to finish deploying the application

The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

2. EasyBuilder Tool Bars
    1. Sensor
    2. System
3. Easy Builder User Interface Application Step 4
    1. Save Job

Follow the steps below to complete the lab exercise:

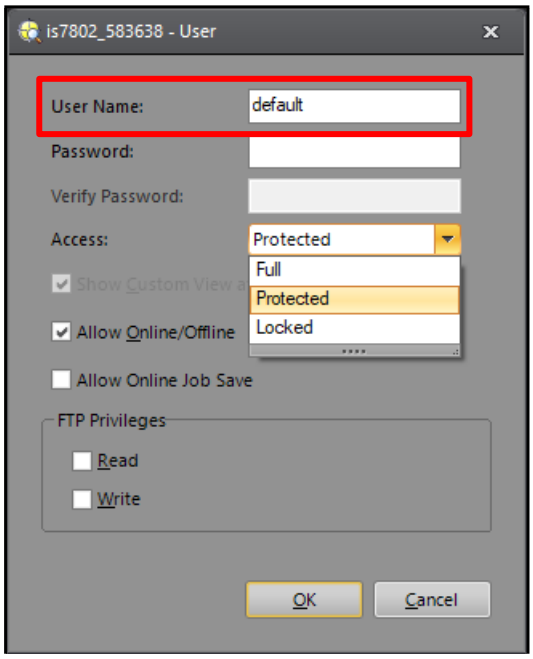
1. Open the **User Access Settings** utility found under the **Sensor** tool bar.



The **User Access Settings** box displays.



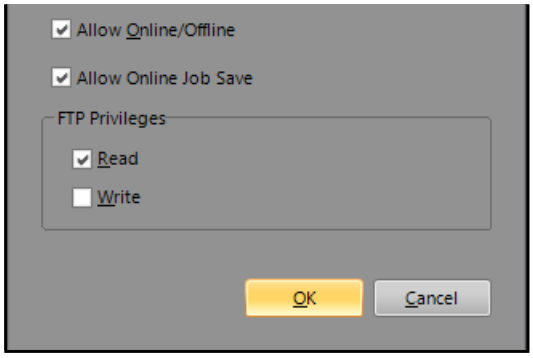
2. Click the **Add** Add... button to create a new user.
   The **New User** box displays.



3. Enter the new username in the **User Name** field.
4. Enter the Password in the **Password** and **Verify Password** fields.

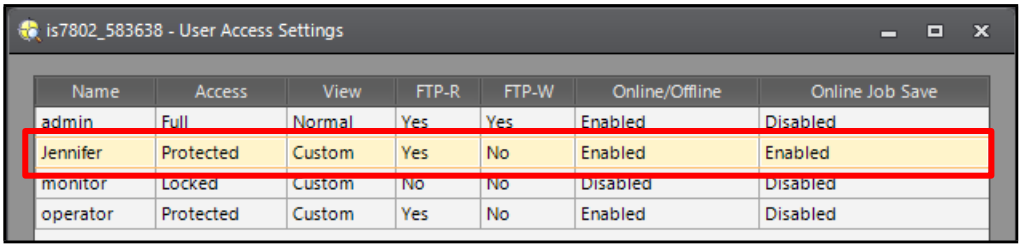
   NOTE: *A password is not required.*
5. Select **Protected** for the Access level.

   NOTE: *The user with Protected access has limited access to the sensor. Protected mode allows you to access Live Video mode, toggle the Online/Offline status of the sensor, open or save jobs, and customize the appearance of the Options dialog. In EasyBuilder view, Protected access allows you to edit tool parameters, but not to add or delete tools.*
6. Keep the **Allow Online/Offline** box checked and check the **Allow Online Job Save** check box.

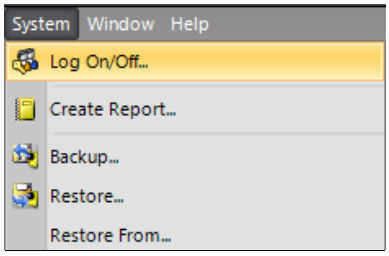


7. Click the **Read** check box for FTP Privileges.
8. Click the **OK** OK button.

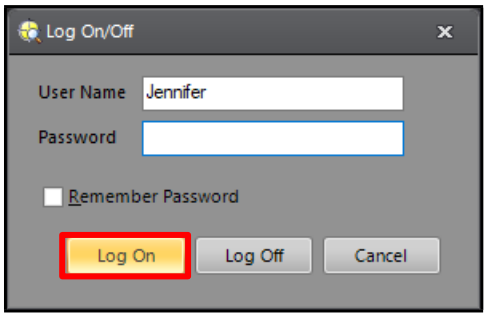
9. The new user is added. Click the **OK**  button to close the User Access Settings dialog.



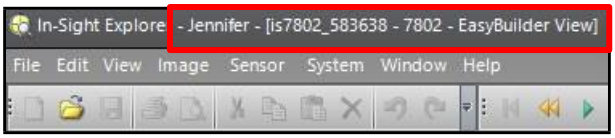
10. Go to the **System** menu and select **Log On/Off**.



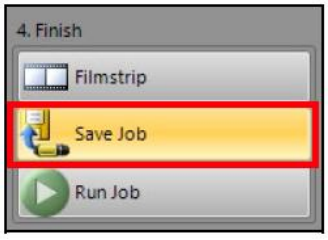
11. Enter the new User Name and Password (if you entered one in the steps above) and click the **Log On**  button.



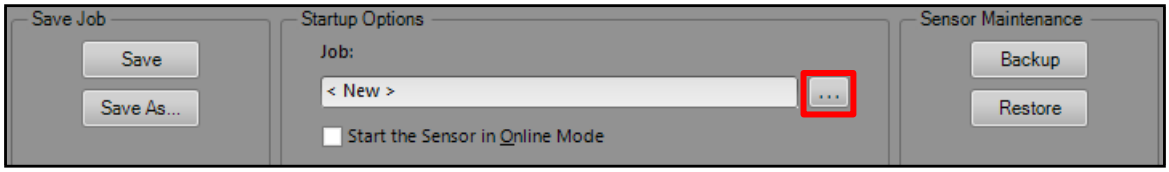
The New User is now logged on the camera.



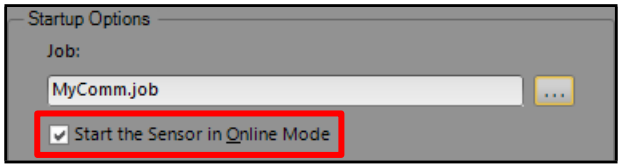
12. Log off the new username and Log back in as **admin**.
13. Click the **Save Job**  button.



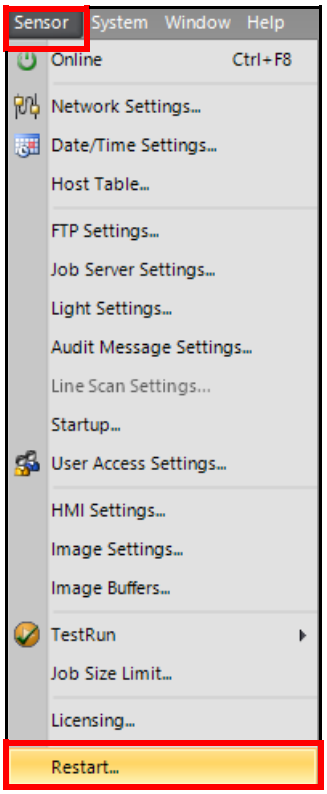
The bottom of the UI – Settings pane will be replaced with the **Save options**.



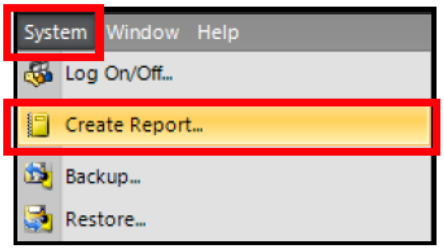
14. Set the **Startup Options** to start with your completed job and click the **Start the Sensor in Online Mode** check box.

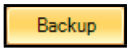


15. Go to the Sensor Menu and select **Restart** to restart your camera to ensure that it works.



16. Go to the System Menu and select **Create Report** to run a report. When the report is complete, review the information to see what is included.



17. Click the **Backup**  button to perform a backup of the system.

18. Once the backup is complete, change the I/O setting and click the **Restore**  button to perform a restore of the system.

19. When the restore is complete, review the **I/O** settings to see if there is any change. This job will not be saved.